

	The intensity alternates from maximum to minimum to maximum 5 times. (Note: when source is very near the detector, the direct path length is zero while the reflected path length is $2 \times 2.5 = 5.0$ cm)	B1
	Max Marks	11

Question	Answer	Marks
4(a)(i)	$\frac{1}{2}(6.2 - 5.8) = 0.2$ m	A1
4(a)(ii)	Since θ is small,	M1

	$\theta = \frac{y}{L}$ $\frac{\Delta\theta}{\theta} = \frac{\Delta y}{y} + \frac{\Delta L}{L}$ percentage uncertainty of separation $y = \frac{0.5}{2.0} \times 100\% = 25\%$ or 30% (1 sf) Average $L = \frac{6.2 + 5.8 + 6.1 + 5.9 + 6.1}{5} = 6.0$ m percentage uncertainty of distance $L = \frac{0.2}{6.0} \times 100\% = 3\%$ The percentage uncertainty of y is almost 10 times that of L, hence I agree that the uncertainty in L can be ignored.	M1 A1
4(a)(iii)	Since θ is small, $\theta = \frac{y}{L} = \frac{0.0020}{6.0} = 0.000333 \text{ rad}$ <u>Method 1:</u> From (c)(ii) the %uncertainty of L can be ignored. $\frac{\Delta\theta}{\theta} = \frac{\Delta y}{y} \Rightarrow \Delta\theta = 0.25 \times 0.000333 = 0.00008 \text{ rad (1 s.f.)}$ Therefore: $0.00033 \pm 0.00008 \text{ rad}$ <u>Method 2:</u> $\theta_{\max} = \frac{y_{\max}}{L_{\min}} = \frac{0.0025}{5.8}$ $\theta_{\min} = \frac{y_{\min}}{L_{\max}} = \frac{0.0015}{6.2}$ $\Delta\theta = \frac{\theta_{\max} - \theta_{\min}}{2} = 0.00009 \text{ rad (1 s.f.)}$ Therefore: $0.00033 \pm 0.00009 \text{ rad}$ Accept both cases where uncertainty in L is included and excluded.	B1 B1 A1
4(b)(i)	The first minimum of pattern 1 coincides with the central maximum for pattern 2 for the 2 patterns to be just distinguishable.	B1 B1
4(b)(ii)	$\theta = \frac{\lambda}{b} = \frac{700 \times 10^{-9}}{3 \times 10^{-3}} = 2 \times 10^{-4} \text{ rad}$	C1 A1
	Max Marks	11

Question	Answer	Marks
5(a)(i)	Kinetic energy, KE is positive Electrical potential energy, EPE is negative	A1 A1
(a)(ii)	Since the electron is orbiting, it means that it is 'confined' to the vicinity of the proton and could not escape on its own. Hence the magnitude of KE is less than that of EPE. Since the total energy of the system E is the sum of its KE and EPE, E must be a negative number.	B1 B1

(b)(i)	If the position of the electron at each location is known exactly, by HUP, there is huge uncertainty in its momentum or vice versa. Hence the path cannot be well defined if either the position or momentum are unknown.	M1 A1
(b)(ii)	The maximum total energy for the system is zero. TE = EPE + KE = 0 $KE = -EPE = -\left(\frac{-e^2}{4\pi\epsilon_0 R}\right) = \frac{e^2}{4\pi\epsilon_0 R}$	B1 B1
	Max Marks	8

Question	Answer	Marks
6(a)	circle unit N	A3